Lab-7

Aim: Interfacing of Temperature and Humidity sensor with embedded target board (Reneses) and control it and on-board LED by using Bluetooth Low Energy (BLE).

Appratus/Software Required: Reneses Development board, Temperature and Humidity sensor

Implementation Steps:

BLE Based Application

Step-1: Create QCstudio Configuration Project

Step-2: Project Name → BLE based Application ↵

Step-3: From QCC Tool PALETTE → Drag "BGK-RA6E2" in workspace

Step-4: From Application configuration: IOT → Other → Free RTOS → Sensor Data Over BLE

Step-5: From QCC Tool PALETTE → connectivity Modules → Drag DA14531 Bluetooth and Connect in PMOD1 of Development Board

Step-6: From QCC Tool PALETTE → SENSORS → Drag HS4001 TEMPERATURE / HUMIDITY Sensor and connect in PMOD 2 of Development Board

Step-7: Generate / Build QCStudio Project. Finally message will appear " QCstudio project build finished successfully "

Step-8: In Debug → bgk_rabe2_freertos.srec → Download file.

Step-9: Connect the Following Hardwares:

- MCU Board > Laptop
- DA14531 Bluetooth > PMOD1
- HS4001 > PMOD2

Step-10 : Launch J-Flash Lite V8.36

- Select Target Device → R7FA6E2BB
- If not listed then select it:

Manufacturer	Device
Renesas	a6e2bb

- Target Interface : SWD
- Speed : 4000 kHz
- Ok ↵
- Browse : .srec file ⟷ Program Device.
- Once Programming completes Press S2 switch ~~on Dev~~ on development board for reset.

Step-11 : • Open " Quick Connect Sandbox Mobile App"

- Select " Bluetooth Low Energy (BLE)
- Device US000-QCS will apper → connect
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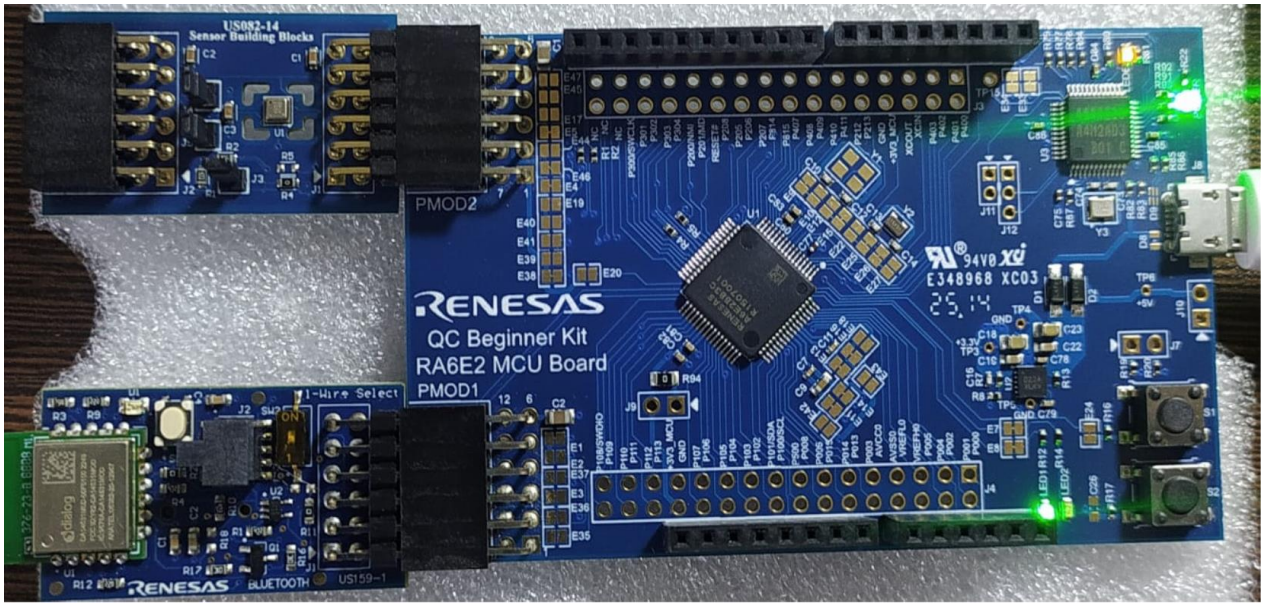

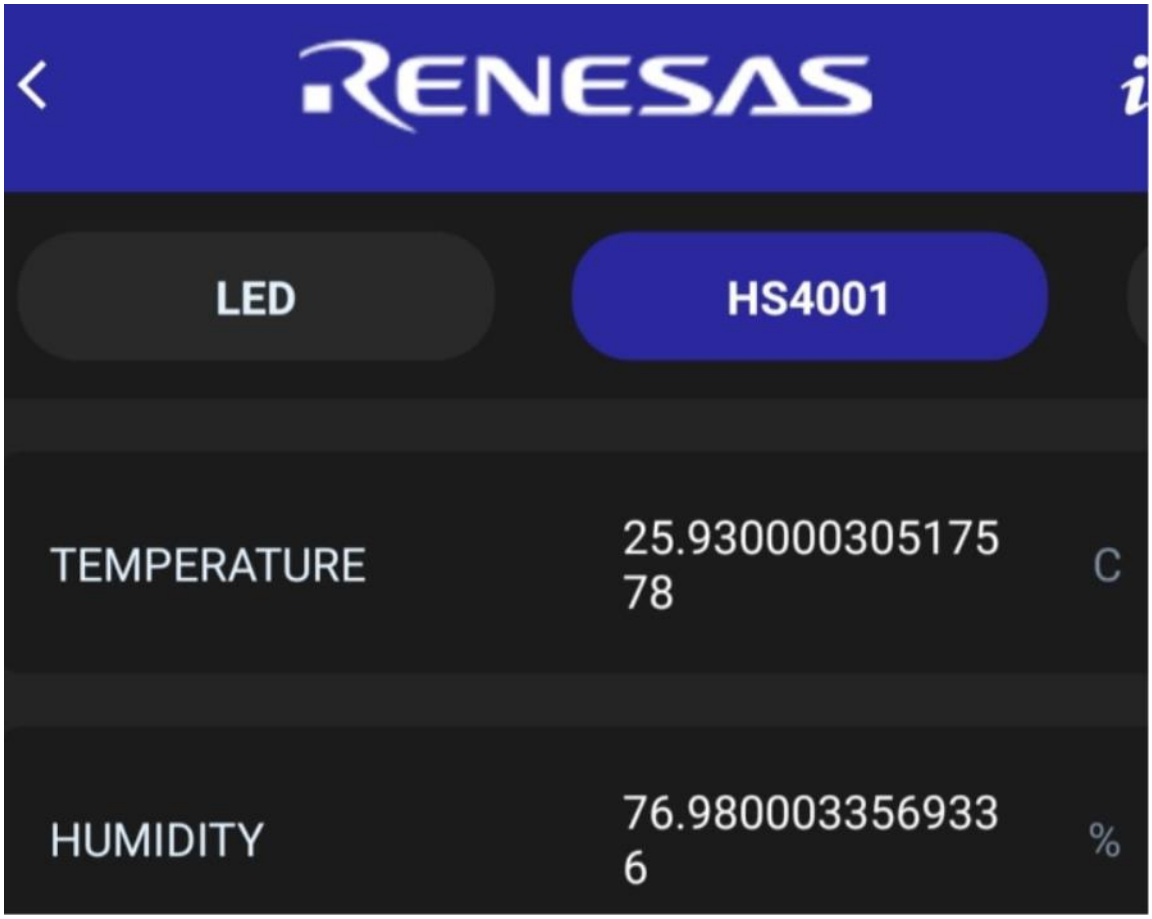

Result: The LED is turned ON and OFF by using blue tooth module and the Temperature and Humidity value is transferred to mobile APP by using blue tooth.